

NovaCart Finance Reporting Agent

Complete Project Summary

Project Overview

I designed and built a fully automated AI-powered finance reporting agent for a fictional e-commerce company called NovaCart, using n8n as the orchestration platform. The agent runs automatically on the 1st of every month, generates a comprehensive CFO-ready report, and delivers it via Gmail and Slack without any human intervention.

Internal RAG Pipeline

The first phase focused on ingesting and querying NovaCart's internal business data. I worked with three proprietary documents: the company's 2024 Annual Report, the full Product Catalog, and three years of weekly SKU-level sales and conversion data with revenue.

I set up a Pinecone vector database index named nova-cart-n8n under the production namespace, configured with 1024 dimensions and cosine similarity using the text-embedding-3-large model. All three documents were chunked at 1000 tokens with a 100-token overlap, resulting in 342 vectors stored in the index.

Three parallel Pinecone RAG nodes query the index simultaneously — one for financial KPIs and revenue data, one for SKU mix and product categories, and one for 3-year weekly sales trends and conversion rates.

External Market Intelligence

The second phase integrated four external research reports as pre-loaded market intelligence signals:

- Deloitte 2026 Semiconductor Industry Outlook
- Deloitte Tech Trends 2026
- McKinsey Technology Trends Outlook 2025
- KPMG 2025 Global Semiconductor Industry Outlook

These four signals provide NovaCart's AI agent with real-world context on semiconductor supply chain risks, consumer electronics demand trends, AI infrastructure costs, tariff impacts on COGS, and the broader technology investment landscape. All four signals run in parallel alongside the three RAG queries, giving the synthesis agent seven concurrent data streams to work with.

AI Synthesis and Delivery

A Python-based Combine Data node merges all seven inputs into a single consolidated context block, which is then passed to a gpt-4o-mini synthesis agent configured with a detailed system prompt. The agent generates a structured 7-section CFO report covering:

- Executive Summary

- Revenue Analysis
- Top 5 SKU Momentum Leaders
- Bottom 5 SKU Underperformers
- External Market Signals
- Risk Register with likelihood and impact ratings
- Prioritized Strategic Recommendations

A DIVERGENCE DETECTED flag automatically surfaces any cases where NovaCart's internal data contradicts external market trends. The completed report is delivered as a branded HTML email to the CFO inbox using Gmail OAuth2, and a summary notification is posted to the #finance-reports channel in the JattaAI Slack workspace. An error handler node logs any failures and sends an alert email if the workflow encounters issues.

Validation Results

In end-to-end testing, the agent successfully retrieved real NovaCart financial data including:

- Total revenue of \$2.84 billion
 - 18.6% year-on-year growth
 - Gross margins of 41.2%
 - NovaPhone X1 identified as top revenue-generating SKU
 - Supply chain disruption flagged as high-likelihood high-impact risk
 - Subscription services expansion recommended as top strategic priority
 - DIVERGENCE DETECTED flag triggered correctly
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Technical Stack

Orchestration: n8n cloud

Vector Database: Pinecone (nova-cart-n8n index, 1024 dimensions, cosine similarity)

Embeddings: OpenAI text-embedding-3-large

AI Synthesis: OpenAI gpt-4o-mini

Email Delivery: Gmail OAuth2

Notifications: Slack OAuth2 — JattaAI workspace, #finance-reports channel

Data Processing: Python code nodes

Schedule: Cron trigger — 1st of every month at 8 AM

Storage: Google Drive OAuth2
